Probability

This Chapter Overview

Solution 2

This chapter is a recap of the concepts you learnt at GCSE.

1: Basic Probability

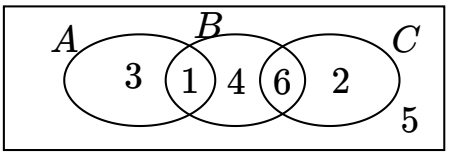
"I throw two fair die. Calculate the probability the sum of the two dice is more than 6."

2: Venn Diagrams

"Out of 50 students, 12 play both piano and drums, 30 play piano and 25 play drums. Find the probability a randomly chosen student plays neither instrument."

3: Mutually Exclusive/Independent Events

Determine whether A and B are independent.



4: Tree Diagrams

"The probability I hit a target is 0.3. If I hit it, the probability I hit again on the next shot is 0.4. If I miss, the probability I hit on the next shot is 0.1. If I shoot 3 times, what's the probability I hit on the first and third shot?"

Probability concepts

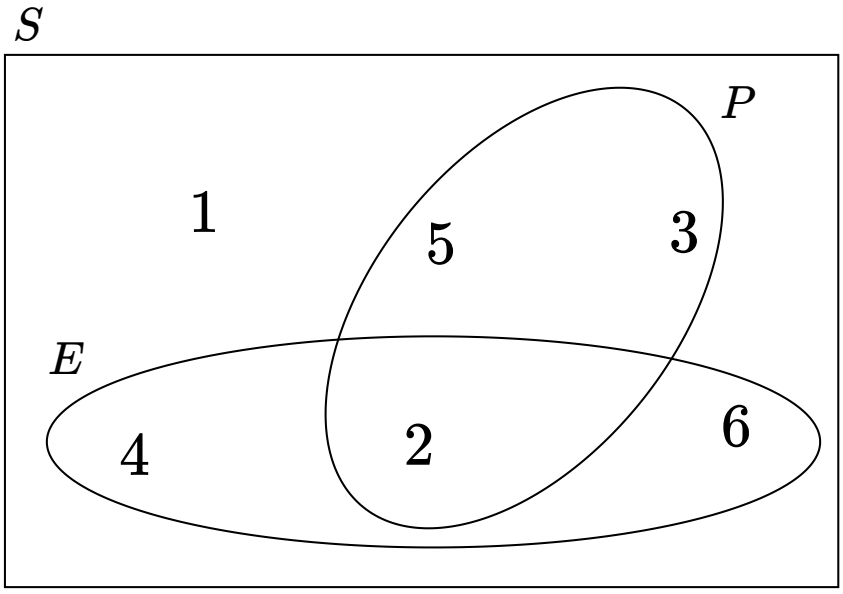
Solution 3



An **experiment** is a repeatable process that gives rise a number a number of **outcomes**.

An **event** is a set of one or more of these outcomes.

(We often use capital letters to represent them)



E = "rolling an even number"
P = "rolling a prime number"

A **sample space** is the set of all possible outcomes.

Because we are dealing with sets, we can use a **Venn diagram**, where

- the numbers are the individual outcomes,
- the sample space is a rectangle and
- the events are sets, each a subset of the sample space.

You do not need to use set notation like $\cap$ and $\cup$ in this module (but ordinarily you would!)

Example

Solution 4

Two fair spinners each have four sectors numbered 1 to 4. The two spinners are spun together and the sum of the numbers indicated on each spinner is recorded.
Find the probability of the spinners indicating a sum of
(a) exactly 5 (b) more than 5

+	Spinner 1: 1	2	3	4
Spinner 2: 1	2	3	4	5
2	3	4	5	6
3	4	5	6	7
4	5	6	7	8

$P(\text{sum} = 5) =$

a $4/16 = 1/4$

$P(\text{sum} > 5) =$

b $6/16 = 3/8$

If the sample space is the amalgamation of two underlying experiments, a table is a helpful way to list the outcomes.

Another Example

Solution 5

The table shows the times taken, in minutes, for a group of students to complete a number puzzle.

Time, t (min)	$5 \le t < 7$	$7 \le t < 9$	$9 \le t < 11$	$11 \le t < 13$	$13 \le t < 15$
Frequency	6	13	12	5	4

A student is chosen at random. Find the probability for a group of students to complete a number puzzle

(a) In under 9 minutes (b) in over 10.5 minutes.

a

$$P(t < 9) = \frac{6+13}{40} = \frac{19}{40}$$

b

$$P(t > 10.5) = \frac{3+5+4}{40} = \frac{3}{10}$$

You're effectively doing **linear interpolation** here: 10.50 to 11 is a quarter of the $9 \le t < 11$ interval, therefore we use a quarter of the 12 people.
This is an estimate because we're assuming the people are equally distributed across the interval.

Exercise 5A – Statistics Year 1 page 71-72

Solution 6

3. The masses of 140 adult Bullmastiffs are recorded in a table. One dog is chosen at random.

(a) Find the probability that its mass is at least 54 kg.

(b) Find the probability that its mass is between 48 and 57 kg.

(c) A randomly chosen Rottweiler has probability 0.54 of weighing under 53 kg. Is a Bullmastiff more or less likely to weigh under 53 kg? State your assumption. Hint: use interpolation.

Mass (kg)	Frequency
$45 \le m < 48$	17
$48 \le m < 51$	25
$51 \le m < 54$	42
$54 \le m < 57$	33
$57 \le m < 60$	21
$60 \le m < 63$	2

a

$$P(m \ge 54) = \frac{33+21+2}{140} = \frac{2}{5}.$$

b

$$P(48 \le m < 57) = \frac{25+42+33}{140} = \frac{5}{7}.$$

c

Let B be the estimated number under 53 kg.
Interpolating within the 51–54 class gives $\frac{53-51}{54-51} = \frac{B-(17+25)}{42}$, so $B = 70$.
$P(m < 53) = 70/140 = 0.5 < 0.54$. A Bullmastiff is less likely to weigh under 53 kg.
We assume masses are uniformly distributed within this class.

Exercise 5A – Statistics Year 1 page 71-72

4. The lengths of 240 koalas were recorded. One koala is chosen at random.

Find the probability that it is:

(a) female;

(b) less than 80 cm long;

(c) male, between 75 cm and 85 cm long.

(d) Koalas under 72 cm are juvenile. Estimate the probability of choosing a juvenile and state one assumption.

Length (cm)	Male	Female
$65 \le l < 70$	4	14
$70 \le l < 75$	20	15
$75 \le l < 80$	24	32
$80 \le l < 85$	47	27
$85 \le l < 90$	31	26

Solution 7

a

$$P(F) = \frac{14+15+32+27+26}{240} = \frac{19}{40}.$$

b

$$P(l < 80) = \frac{4+14+20+15+24+32}{240} = \frac{109}{240}.$$

c

$$P(M \cap \{75 \le l < 85\}) = \frac{24+47}{240} = \frac{71}{240}.$$

d

Interpolate for lengths between 70 and 75 cm.

Male juveniles: $4 + \frac{72-70}{75-70}(20) = 12$.

Female juveniles: $14 + \frac{2}{5}(15) = 20$.

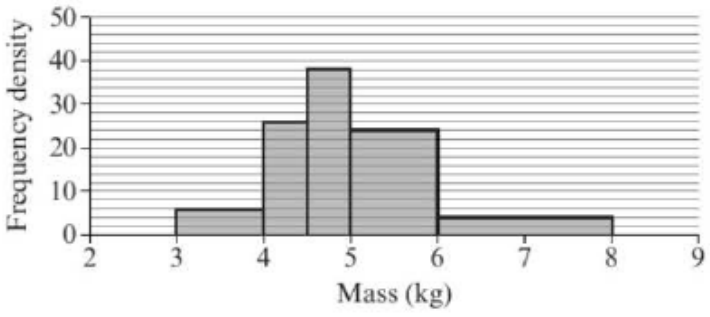
$P(\text{juvenile}) \approx \frac{12+20}{240} = \frac{2}{15}$. Assume lengths are uniform within that class.

Exercise 5A – Statistics Year 1 page 71-72

5. The histogram shows the distribution of masses of 70 adult cats.

(a) Find the probability that a randomly chosen cat has mass greater than 5 kg. (2 marks)

(b) Estimate the probability that a randomly chosen cat has mass less than 6.5 kg. (3 marks)



Solution 8

a

$$P(m > 5) = \frac{1(24)+2(4)}{70} = \frac{16}{35}.$$

b

First estimate the probability of a mass greater than 6.5 kg. The final class is two units wide and three quarters of it is above 6.5 kg.

$P(m > 6.5) = \frac{(3/4)(2)(4)}{70} = \frac{3}{35}$.

$P(m < 6.5) = 1 - \frac{3}{35} = \frac{32}{35}$.

Interpolation assumes a continuous distribution; the probability of any single exact mass is negligible.

Extension

Solution 9

[Challenge p72] Samira chooses one card at random from each group. Group A has 2, 7, 5; group B has 4 and x. She records their product Y. Given x is an integer and $P(Y \text{ is even}) = P(Y \geq 20)$, find the possible values of x.

Solution

If x is even, every product is even but $2 \times 4 = 8 < 20$, so the two probabilities cannot be equal.

For odd x, four products are even: $8, 28, 20, 2x$. Four products must therefore be at least 20. Two are 28 and 20; precisely two of $2x, 7x, 5x$ must qualify.

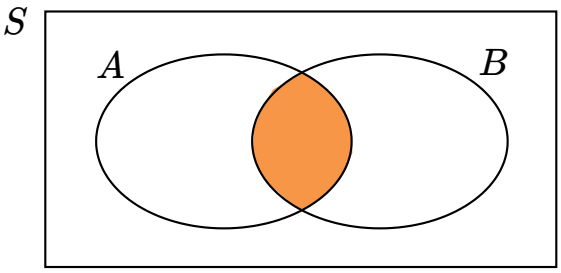
Hence $2x < 20$ and $5x \geq 20$, so $4 \leq x < 10$. The possible odd integers are $5, 7, 9$.

B × A	2	7	5
4	8	28	20
x	2x	7x	5x

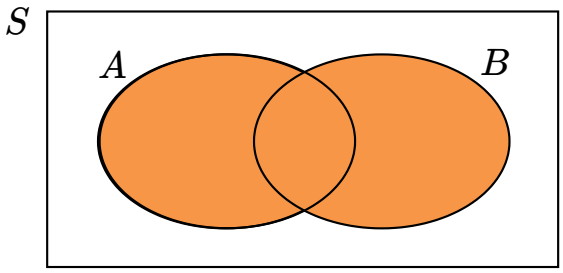
Venn Diagrams

Solution 10

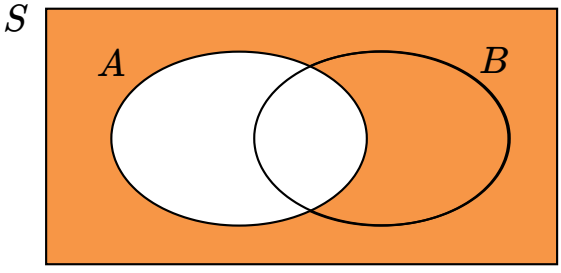
Venn Diagrams allow us to combine events, e.g. "A happened **and** B happened".



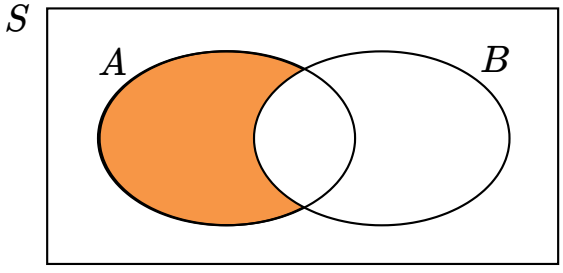
The event "A **and** B"
Known as the **intersection** of A and B.



The event "A **or** B"
Known as the **union** of A and B.



The event "not A"
Known as the complement of A.



These can be combined,
e.g. "A and not B".

Example involving probabilities

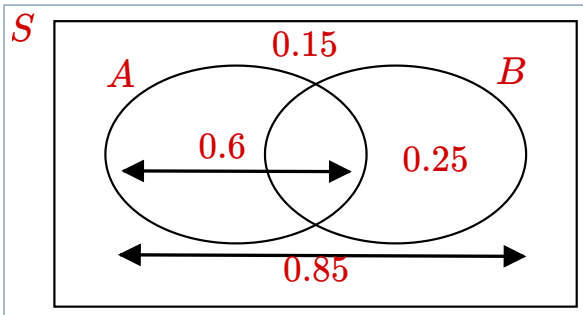
Solution 11

We can either put frequencies or probabilities into the Venn Diagram.

Given that $P(A) = 0.6$ and $P(A or B) = 0.85$, find the probability of:

a) $P(not A and B)$

b) $P(neither A nor B)$



$P(not A and B) = \mathbf{0.85 - 0.6 = 0.25}$

$P(neither A nor B) = \mathbf{1 - 0.85 = 0.15}$

Example involving frequencies

Solution 12

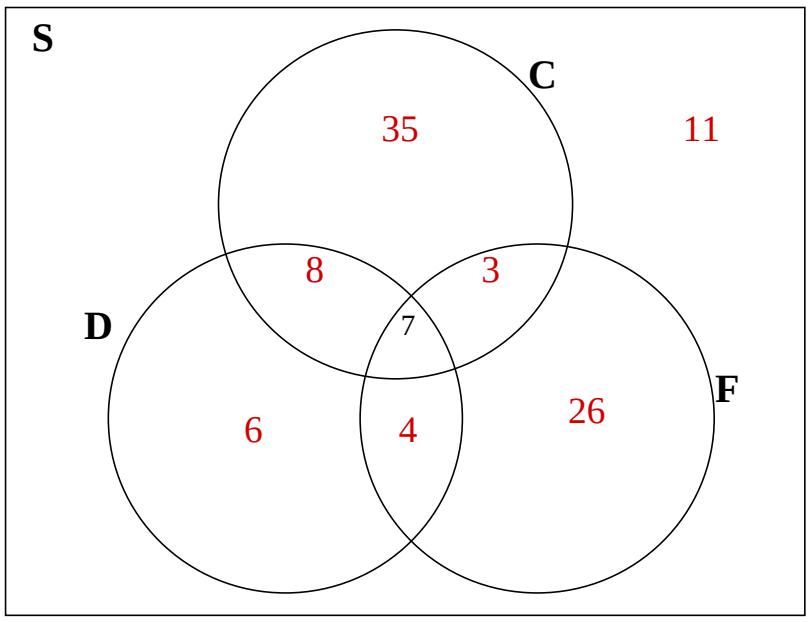
A vet surveys 100 of her clients. She finds that
25 own dogs, 15 own dogs and cats, 11 own dogs and tropical fish, 53 own cats, 10 own cats and tropical fish, 7 own dogs, cats and tropical fish, 40 own tropical fish.

Fill in this Venn Diagram, and hence answer the following questions:

a) $P(owns dog only)$

b) $P(does not own tropical fish)$

c) $P(does not own dogs, \ \ cats, or tropical fish)$



Tip: Start from the centre frequency and work your way outwards using subtraction.

a) $\frac{6}{100} = \frac{3}{50}$

b) $\frac{60}{100} = \frac{3}{5}$

c) $\frac{11}{100}$

Test Your Understanding

Solution 13

Jan 2012 Q6. Of 100 people, 65 run, 48 swim, 60 cycle, 40 run and swim, 30 swim and cycle, 35 run and cycle and 25 do all three.

(a) Draw a Venn diagram. (4 marks) Tip: include the box.

Find the probability that a randomly chosen person:

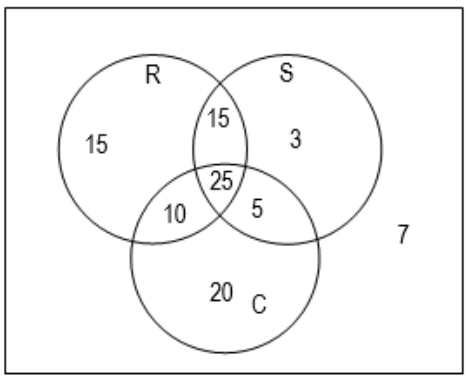
(b) takes none of these types of exercise (2 marks);

(c) swims but does not run (2 marks);

(d) takes at least two types of exercise (2 marks).

(e) Given that Jason runs, find the probability that he swims but does not cycle. (3 marks)

a



b $P(\text{none}) = 7/100 = 0.07.$

c $P(S \cap R') = (3+5)/100 = 0.08.$

d $P(\text{at least two}) = (25+15+10+5)/100 = 0.55.$

e

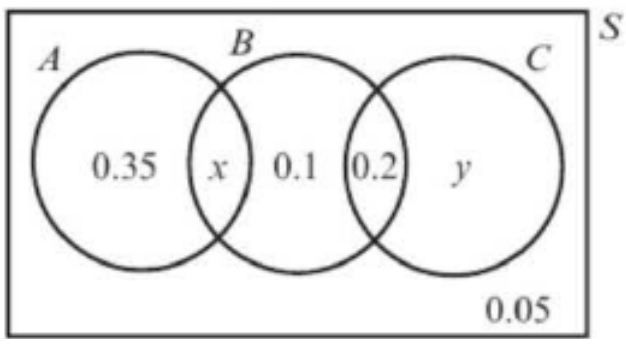
$$P(S \cap C' \mid R) = \frac{P(S \cap C' \cap R)}{P(R)} = \frac{15}{65} = \frac{3}{13}.$$

Exercise 5B – Statistics Year 1 page 74-75

Solution 14

5. In a doctor's waiting room a patient reads Hiya with probability 0.6, Dakar with probability 0.4, and either or both with probability 0.7. Find the probability of reading (a) both magazines (2 marks), (b) Hiya only (2 marks).

6. The diagram shows participation in archery (A), badminton (B) and croquet (C). Given $P(B) = 0.45$, find (a) x (1 mark) and (b) y (2 marks).



5a

$P(H \cap D) = P(H) + P(D) - P(H \cup D) = 0.6 + 0.4 - 0.7 = 0.3.$

5b

$P(H \text{ only}) = 0.6 - 0.3 = 0.3.$

6a

$P(B) = x + 0.1 + 0.2 = 0.45$, so $x = 0.15.$

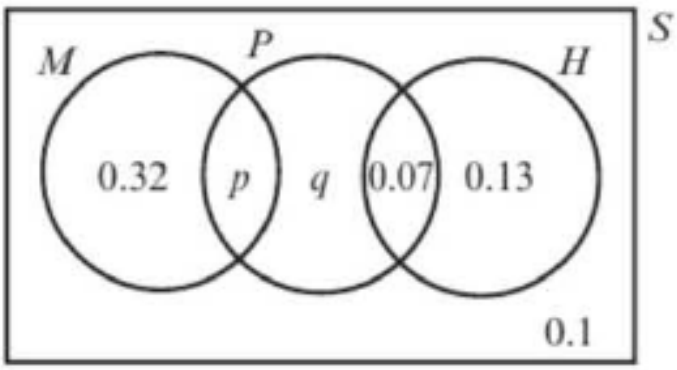
6b

$y = 1 - (0.35 + 0.15 + 0.1 + 0.2 + 0.05) = 0.15.$

Exercise 5B – Statistics Year 1 page 74-75 Solution 15

7. The diagram shows which subjects sixth-form students study. M is mathematics, P physics, H history.

Given $P(M) = P(P)$, find p and q. (4 marks)



7

$P(M) = 0.32 + p$, and $P(P) = p + q + 0.07$.

As the probabilities are equal, $0.32 + p = p + q + 0.07$, giving $q = 0.25$.

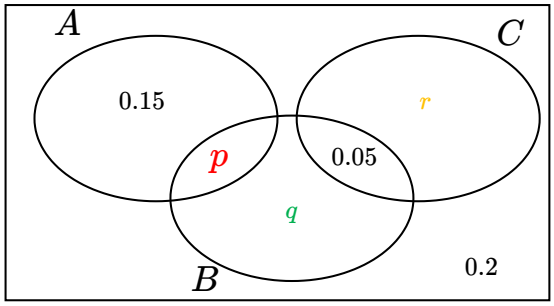
$p = 1 - (0.32 + 0.25 + 0.07 + 0.13 + 0.1) = 0.13$.

$(p, q) = (0.13, 0.25)$.

Extension Solution 16

[Challenge p75] The Venn Diagram shows the probabilities of a group of children liking three types of sweet.

Given that $\mathrm{P}(B) = 2\mathrm{P}(A)$ and that $\mathrm{P}(notC) = 0.83$, find the values of p, q, and r.



$P(B) = p + q + 0.05$, $P(A) = 0.15 + p$. Since $P(B) = 2P(A)$, $q = 0.25 + p$.

$P(C') = 0.83$ gives $0.15 + p + q + 0.2 = 0.83$, so $p + q = 0.48$.

$p + (0.25 + p) = 0.48$, hence $p = 0.115$ and $q = 0.365$.

$P(C) = 0.17 = r + 0.05$, so $r = 0.12$.

Mutually Exclusive Events

Solution 17

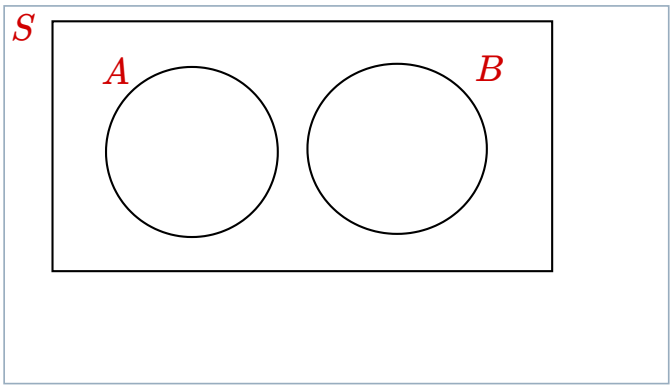
If two events are mutually exclusive, they cannot happen at the same time.

If A and B are mutually exclusive, then

$P(A \cap B) =$ 0

$P(A \cup B) =$ $P(A) + P(B)$

The Venn diagram would look like this.



Mutually exclusive events have no outcomes in common. Their regions cannot overlap.

Independent Events

Solution 18

If two events are independent,

whether one happens does not affect the probability of the other happening.

If A and B are independent, then

$P(A \cap B) =$ $P(A) \times P(B)$

Note: Independence does not affect how the circles interact in a Venn Diagram.

Example 1 2 3 4

1 I pick one of 1, 2, 3, 4 at random. What is the probability that:

I pick a multiple of 2? 1/2

I pick a multiple of 4? 1/4

2 Explain conceptually why the events are not independent.

A multiple of 4 must also be a multiple of 2. If it is not a multiple of 4 it may or may not be a multiple of 2. Knowing one outcome changes the probability of the other.

3 Show that the events are not independent.

$P(A)P(B) = \frac{1}{2} \times \frac{1}{4} = \frac{1}{8}$.
$P(A \cap B) = \frac{1}{4}$. Since $\frac{1}{8} \neq \frac{1}{4}$, the events are not independent.

This is a common exam question. Either show that $P(AandB) = P(A) \times P(B)$
or that $P(AandB) \neq P(A) \times P(B)$

Further Examples

Solution 19

[Textbook] Events A and B are mutually exclusive and $P(A) = 0.2$ and $P(B) = 0.4$.

a) Find $P(A or B)$

b) Find $P(A but not B)$

c) Find $P(neither A nor B)$

a $P(A \cup B) = 0.2 + 0.4 = 0.6$

b $P(A \cap B') = P(A) = 0.2$

c $P((A \cup B)') = 1 - 0.2 - 0.4 = 0.4$

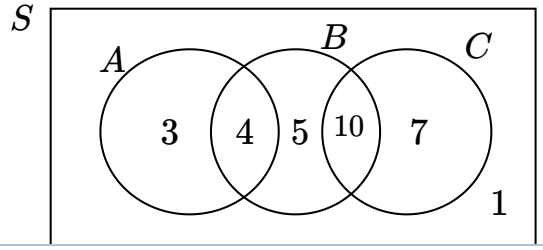
[Textbook] Events A and B are independent and $P(A) = \frac{1}{3}$ and $P(B) = \frac{1}{5}$. Find $P(A and B)$.

$P(A and B) = \frac{1}{3} \times \frac{1}{5} = \frac{1}{15}$

[Textbook] The Venn diagram shows the number of students in a particular class who watch any of three popular TV programmes.

a) Find the probability that a student chosen at random watches B or C or both.

b) Determine whether watching A and watching B are statistically independent.



a $P(watches B or C) = \frac{26}{30}$

b

$P(A) = \frac{7}{30}$ $P(B) = \frac{19}{30}$

$P(A and B) = \frac{4}{30}$

$P(A) \times P(B) = \frac{7}{30} \times \frac{19}{30} = \frac{133}{900} \neq \frac{4}{30}$

$\therefore$ not independent.

Test Your Understanding

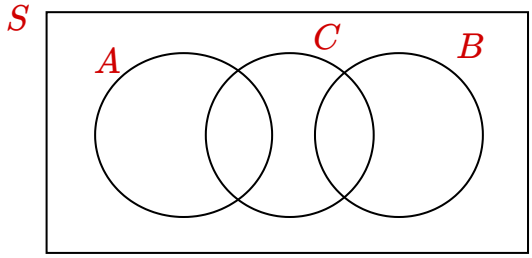
Solution 20

There are three events A, B, C. The events A and B are mutually exclusive.

a) Draw a Venn diagram which represents this information.

b) If $P(A) = 0.1$ and $P(B) = 0.6$, determine $P(neither A nor B)$

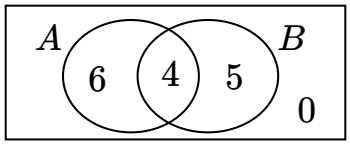
a



Note that C may occur with A or with B, as we're not told otherwise.

b $\mathbf{P(neither A nor B) = 1 - 0.1 - 0.6 = 0.3}$

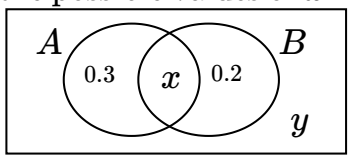
The Venn diagram shows the number of people who like each of two different colours. Determine if A and B are independent.



$\mathbf{P(A) = \frac{10}{15} = \frac{2}{3}}$ $\mathbf{P(B) = \frac{9}{15} = \frac{3}{5}}$, $\mathbf{P(A and B) = \frac{4}{15}}$

$\mathbf{P(A) \times P(B) = \frac{2}{3} \times \frac{3}{5} = \frac{6}{15} \neq \frac{4}{15}}$ $\mathbf{\therefore}$ **not independent.**

The Venn diagram shows the probability of each event. Given that A and B are independent, determine the possible values of x.



$\mathbf{P(A and B) = P(A) \times P(B)}$

$\mathbf{x = (0.3 + x)(0.2 + x)}$

Solving the quadratic, $\mathbf{x = 0.2 or x = 0.3}$

Exercise 5C – Statistics Year 1 page 77-78

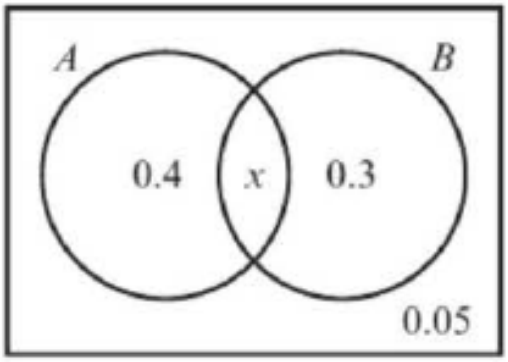
Solution 21

6. The Venn diagram shows the probabilities that students like pasta (A) or pizza (B).

(a) Write down the value of x. (1 mark)

(b) Determine whether liking pasta and liking pizza are independent. (3 marks)

8. $P(W) = 0.5$, $P(W \cap X') = 0.25$ and $P((W \cup X)') = 0.3$. Are W and X independent? Give a reason. (3 marks)



6a

$0.4 + x + 0.3 + 0.05 = 1$, so $x = 0.25$.

6b

$P(A \cap B) = 0.25$ but $P(A)P(B) = 0.65(0.55) = 0.3575$. The events are not independent.

8

$P(W \cap X) = P(W) - P(W \cap X') = 0.25$.

$P(X) = 1 - (0.25 + 0.3) = 0.45$.

$P(W)P(X) = 0.5(0.45) = 0.225 \neq 0.25$, so the events are not independent.

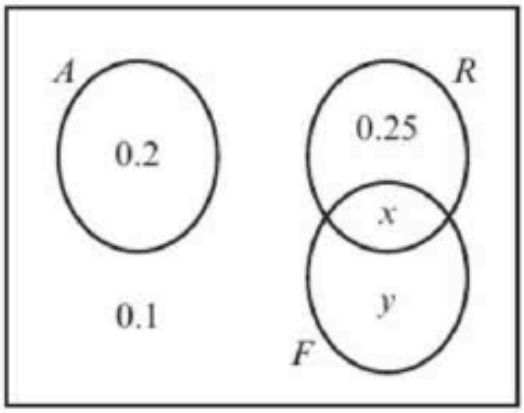
Exercise 5C – Statistics Year 1 page 77-78

Solution 22

9. A social club runs an archery competition (A), raffle (R) and fun run (F). The Venn diagram shows members' participation. The probability of taking part in archery or the raffle is 0.6.

(a) Find x and y. (2 marks)

(b) Show that R and F are not independent. (3 marks)



a

A and R are mutually exclusive, so $P(A \cup R) = 0.2 + 0.25 + x = 0.6$. Thus $x = 0.15$.

$y = 1 - (0.2 + 0.25 + 0.15 + 0.1) = 0.30$.

b

$P(R \cap F) = 0.15$.

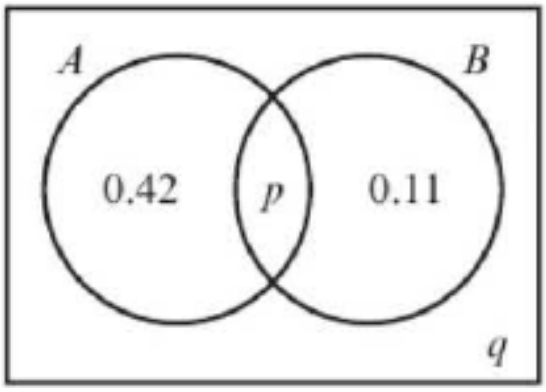
$P(R)P(F) = 0.4(0.45) = 0.18 \neq 0.15$.

Therefore the events are not independent.

Exercise 5C – Statistics Year 1 page 77-78

Solution 23

10. Given that A and B are independent, find the two possible pairs of values for p and q.



10

$P(A \cap B) = p$. Independence gives $p = (0.42 + p)(p + 0.11)$.

$p^2 - 0.47p + 0.0462 = 0$.

$p = \frac{0.47 \pm \sqrt{0.47^2 - 4(0.0462)}}{2} = \frac{0.47 \pm 0.19}{2}$.

$p = 0.33$ or 0.14. Since $q = 1 - (0.42 + p + 0.11)$, the possible pairs (p, q) are $(0.14, 0.33)$ and $(0.33, 0.14)$.

Extension

Solution 24

If A and B are independent, prove that:
(a) A and the complement of B are independent;
(b) the complements of A and B are independent.

a

Let $P(A) = p$, $P(B) = q$. Then $P(A \cap B) = pq$.

$P(A \cap B') = P(A) - P(A \cap B) = p - pq$.

This is $p(1 - q) = P(A)P(B')$, proving independence.

b

$P(A' \cap B') = 1 - P(A \cup B)$

$= 1 - P(A) - P(B) + P(A \cap B)$

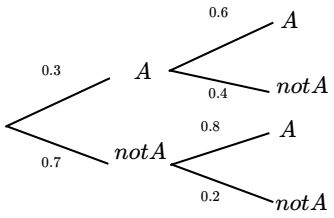
$= 1 - p - q + pq = (1 - p)(1 - q)$.

This equals $P(A')P(B')$, proving independence.

Tree Diagrams — Solution 25

At GCSE we saw that tree diagrams were an effective way of showing the outcome of two events which happen **in succession**.

(Personal opinion however is that their use is easily avoidable)



There are 3 yellow and 2 green counters in a bag. I take two counters at random. Determine the probability that:

a) They are of the same colour.

b) They are of different colours.

a

$YY: \frac{3}{5} \times \frac{2}{4} = \frac{6}{20}$

$GG: \frac{2}{5} \times \frac{1}{4} = \frac{2}{20}$

$P(same) = \frac{6}{20} + \frac{2}{20} = \frac{8}{20} = \frac{2}{5}$

list out the matching sequences of outcomes first, then find the probability for each.

b

$GY: \frac{2}{5} \times \frac{3}{4} = \frac{6}{20}$

$YG: \frac{3}{5} \times \frac{2}{4} = \frac{6}{20}$

$P(different) = \frac{6}{20} + \frac{6}{20} = \frac{3}{5}$

(Or we could have used our answer to (a): $1 - \frac{2}{5} = \frac{3}{5}$)

Note that if the outcomes are the same but reordered, the probability will be the same. So we could have quickly done $\frac{2}{5} \times \frac{3}{4} \times 2$

The probability I hit a target on each shot is 0.3. I keep firing until I hit the target. Determine the probability I hit the target on the 5th shot.

$0.7^4 \times 0.3 = 0.07203$

Exercise 5D – Statistics Year 1 page 79-80 — Solution 26

4. Mr Dixon scores par or under on the first hole with probability 0.7. Given par or under on the first hole, the probability of par or under on the second is 0.8. Otherwise it is 0.4.

(a) Draw a tree diagram. (3 marks)

(b) Are scoring par or under on each of the two holes independent events? (1 mark)

(c) Find the probability of par or under on exactly one hole. (3 marks)

b

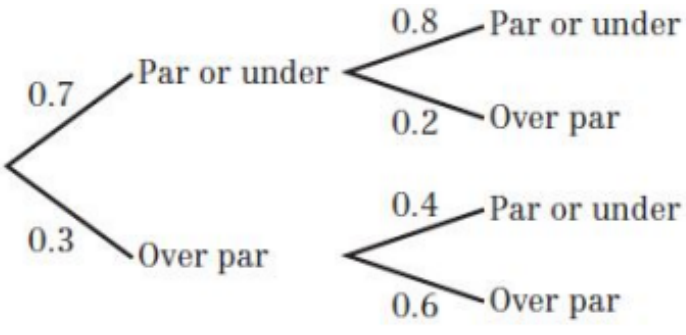
No. The probability on the second hole depends on the result at the first.

c

Par or under then over, or over then par or under:

$0.7(0.2) + 0.3(0.4) = 0.26.$

a

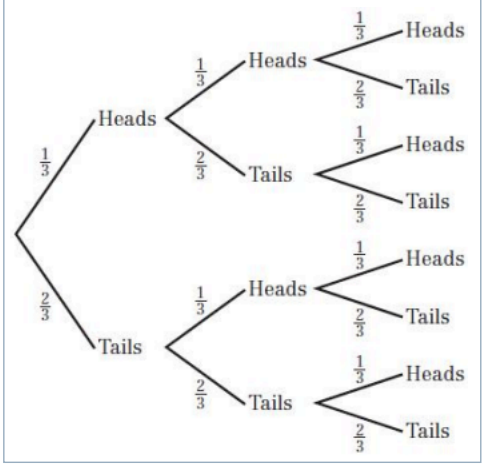


Exercise 5D – Statistics Year 1 page 79-80

Solution 27

5. A biased coin with $P(H) = 1/3$ is tossed three times.
(a) Draw a tree diagram. (3 marks)
(b) Find the probability of three heads. (1 mark)
(c) Find the probability of exactly one head. (2 marks)
(d) Repeat the experiment for a second trial. Find the probability of obtaining three heads or three tails in both trials. (3 marks)

a



b

$P(HHH) = (1/3)^3 = 1/27.$

c

The three orders are HTT, THT, TTH. Each has probability $(1/3)(2/3)^2 = 4/27$. Thus $P(\text{one head}) = 4/9$.

d

$P(HHH \text{ or } TTT) = (1/3)^3 + (2/3)^3 = 1/3.$
For both trials the probability is $(1/3)^2 = 1/9$.

Exercise 5D – Statistics Year 1 page 79-80

Solution 28

6. A bag contains 13 tokens: 4 blue, 3 red and 6 yellow. Draw two without replacement.
(a) Find the probability that both are yellow. (2 marks)
Draw a third token.
(b) Find the probability it is yellow given that the first two were yellow. (1 mark)
(c) Find the probability all three tokens are different colours. (4 marks)

a

$$P(YY) = \frac{6}{13} \times \frac{5}{12} = \frac{5}{26}.$$

b

$$P(Y_3 \mid Y_1 \cap Y_2) = \frac{4}{11}.$$

c

The six orders are BRY, BYR, RYB, RBY, YBR and YRB. Each has probability $\frac{4\times3\times6}{13\times12\times11}$.
Thus $P(\text{all different}) = 6\frac{4\times3\times6}{13\times12\times11} = \frac{432}{1716} = \frac{36}{143}$.